

Air University, Islamabad
Department of Creative Technologies



Course: Digital Logical Design – Lab

BSSE – II – A

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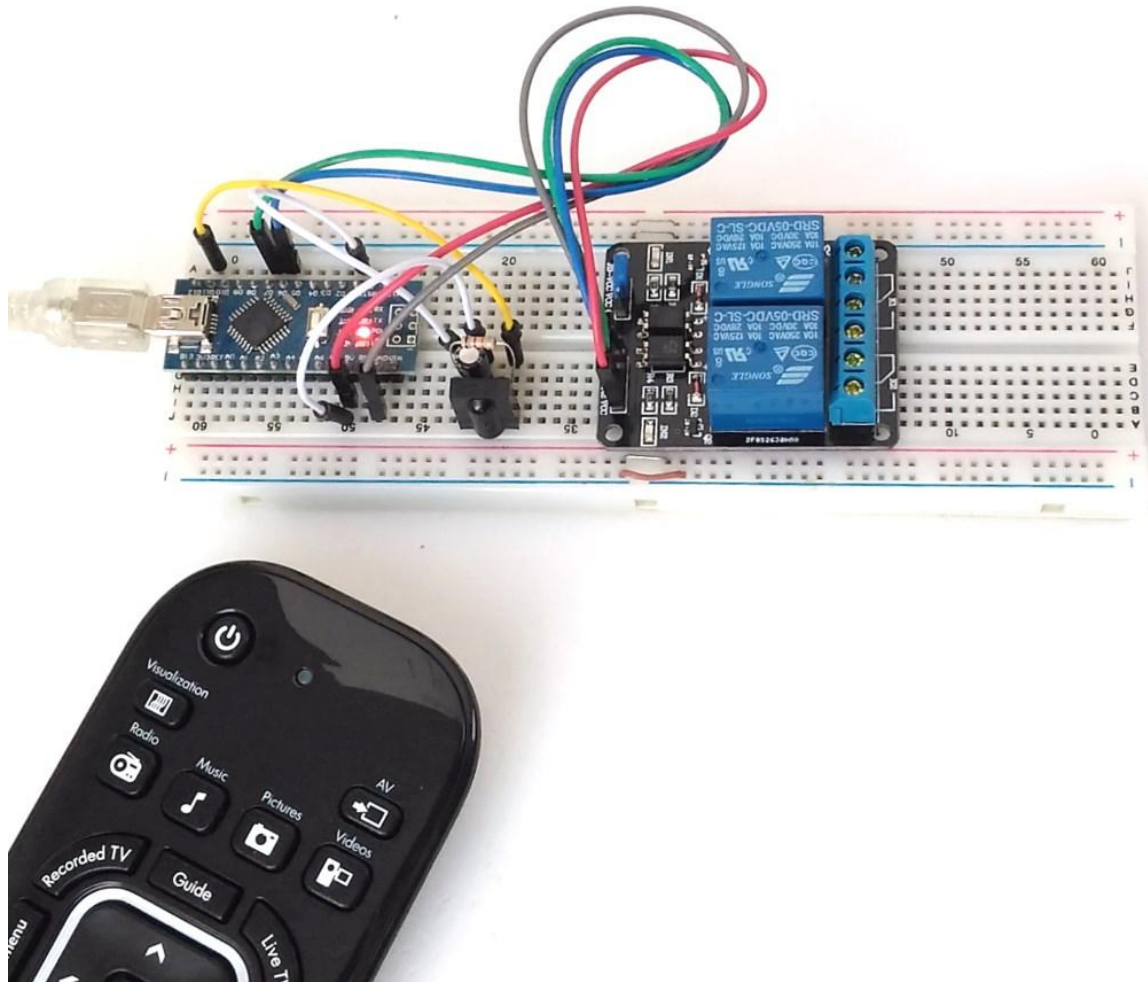
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Project title: Controlling Relay Switches with an Infrared Remote



Project Objective

This project aims to design and implement a system to control relay switches using an infrared (IR) remote. The main goal is to enable wireless control of electrical appliances. The project demonstrates how IR technology can be effectively used for home automation applications to improve user convenience and safety by eliminating the need for manual switching.

Introduction

Infrared (IR) communication is often used in devices like TVs and audio systems for easy remote control. This project uses an IR remote to control relay switches through an Arduino Nano.

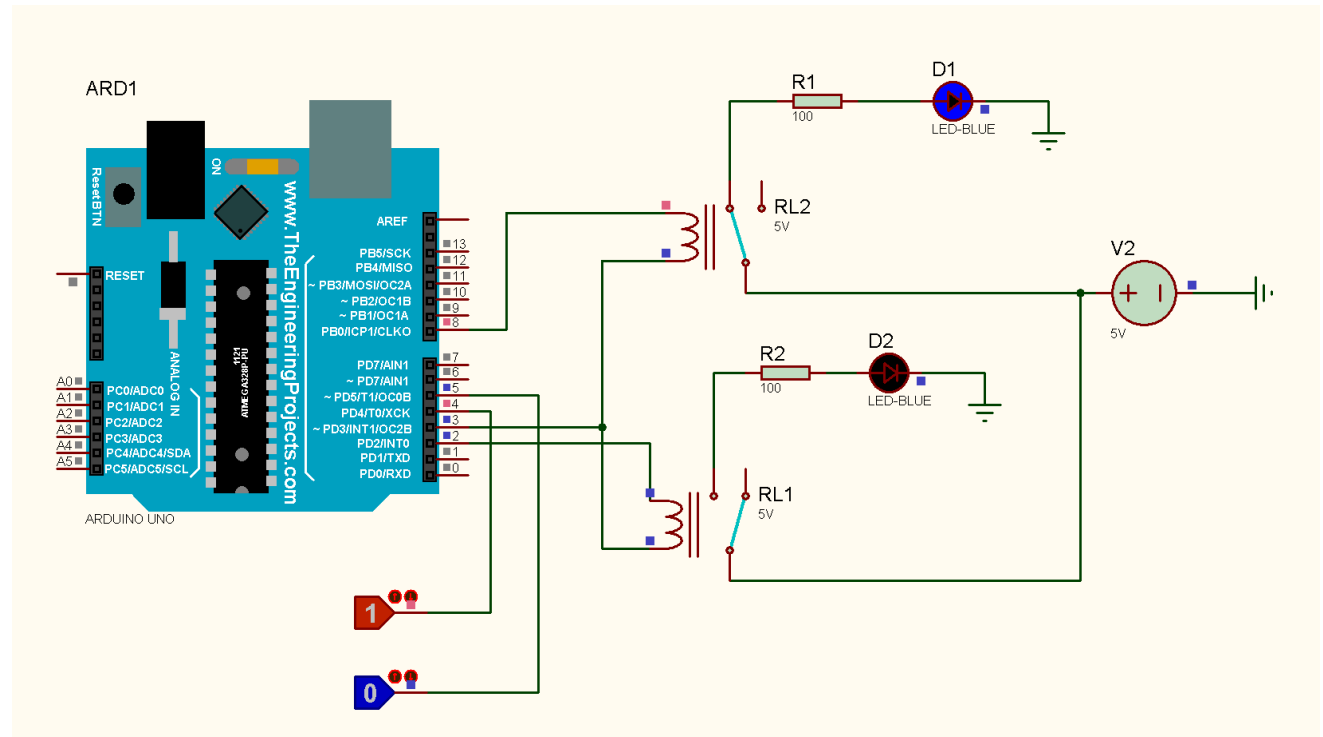
These relays can turn on and off devices like lights, fans, or other appliances. Many people want an easier way to manage their home appliances without moving around every time. This project addresses that need by offering a simple and affordable solution. It does not require internet, so it works even without Wi-Fi or Bluetooth. Unlike other designs that need extra apps or complex setups, this project is easy to build and understand. The low cost and easy-to-find parts make it very practical for home use. Overall, this project helps save time, money, and effort in daily life.

Experimental Work:

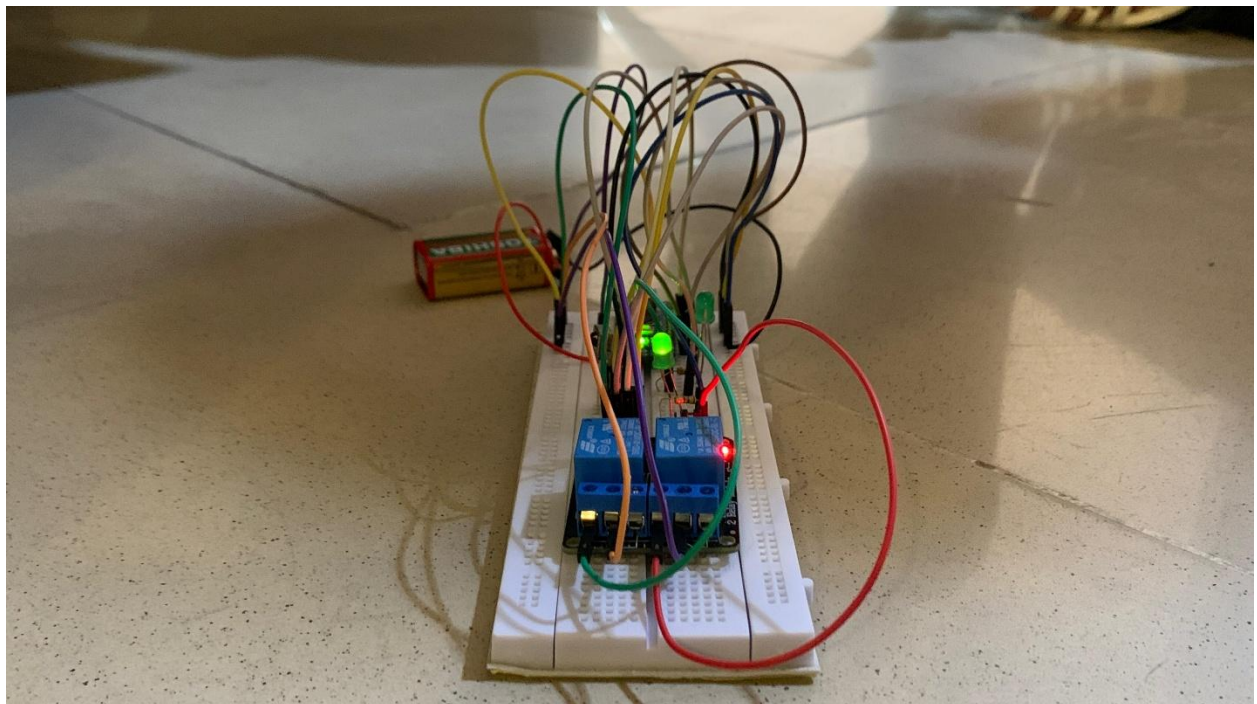
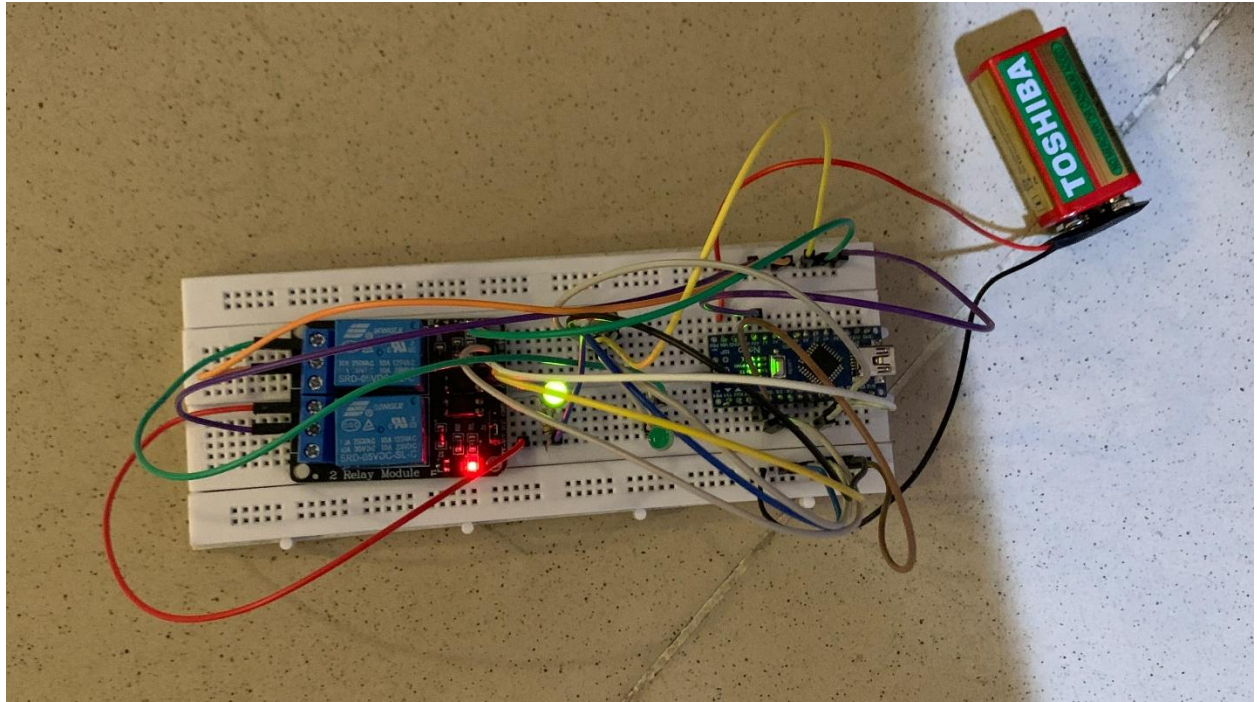
Components Required:

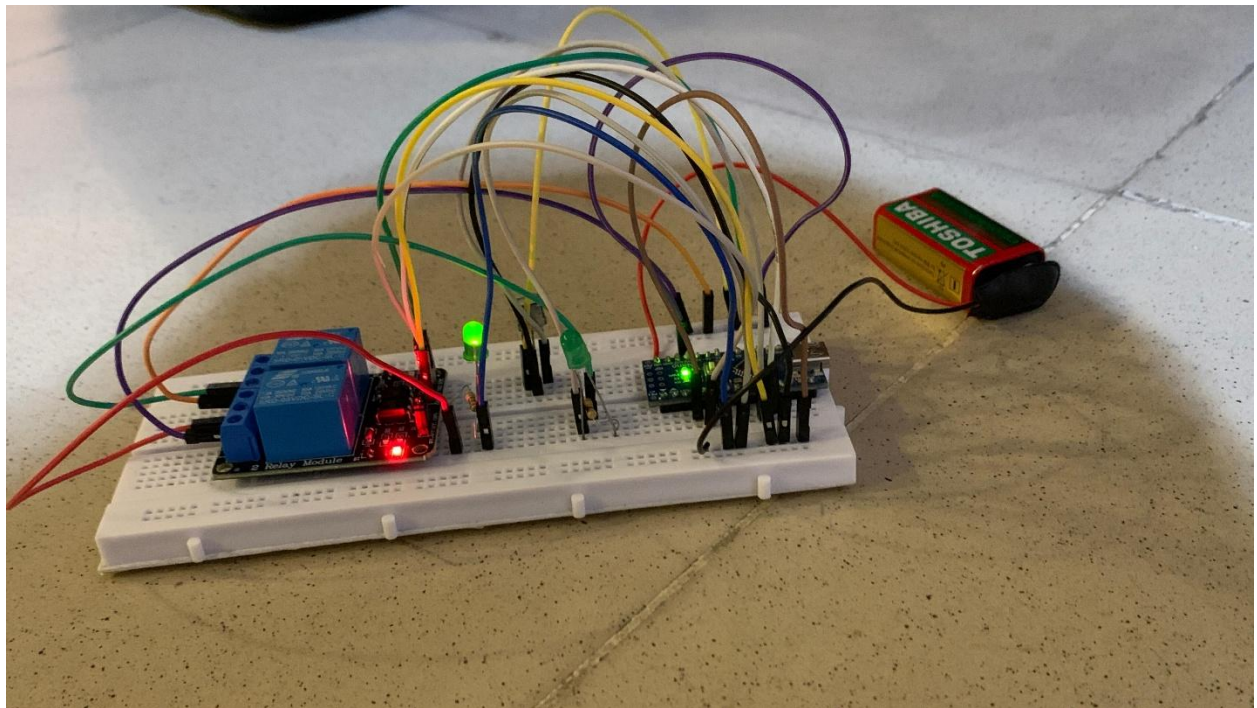
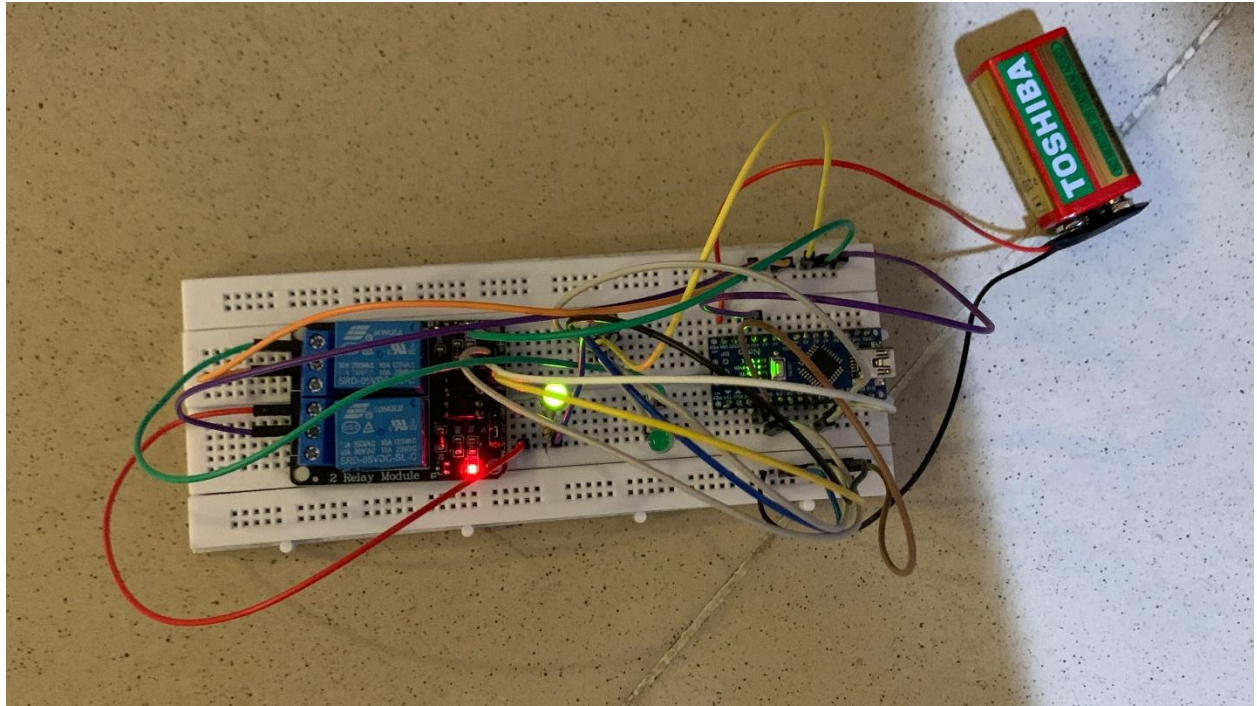
- [Arduino Nano microcontroller](#)
- [TSOP1838 IR Receiver](#)
- [Relay Module \(2 Channel\)](#)
- [9V Battery](#)
- [9V Battery Connector](#)
- [Jumper Wires](#)
- [Breadboard](#)
- [Resistors](#)
- [Two LEDs for testing](#)
- [Bulb holder](#)
- [Zero-watt bulb](#)

Circuit Design:



Project images:





Project image (after amendments)

- This circuit was modified by incorporating an LED holder along with a zero-watt bulb, which serves as an indicator or light source. These enhancements were made with the commercial intent of utilizing the circuit in home automation applications, such as automated lighting control.



Shot video link:

https://youtu.be/7GQMGZqNYjo?si=EsasQL_tDSiqTHOI

Commercialization ideas:

1. Home Automation

- The IR-controlled relay system can be integrated into homes for convenient appliance control.
- Users can operate lights, fans, and entertainment systems from their existing IR remotes.
- A compact version of the system could be installed in wall switches or power sockets for easy access.

2. Event & Stage Lighting

- This technology could be used in fashion show ramps to control lighting sequences.
- Remote switching of stage lights can enhance performance and create dynamic visual effects.
- It can be scaled up to work with multiple relays for synchronized lighting in concerts, theaters, and exhibitions.

3. Smart Office & Workspaces

- Offices can use IR relay switches to automate lights and electronic devices.
- Conference rooms can have remote-controlled projectors, screens, and lighting systems.
- Employee desks could have personalized remote-controlled power outlets for convenience.

4. Industrial Applications

- Factories can deploy IR relay switches to control machinery and conveyor belts remotely.
- Warehouse lighting systems can be automated without requiring expensive wireless setups.
- Small-scale automation systems can benefit from cost-effective IR relay control.

Estimated cost (per unit):

- Microcontroller (Arduino or custom chip): Rs 1500–Rs 1800
- IR Receiver & Remote: Rs 400–Rs 500
- Relay Module: Rs 800–Rs 1000
- Housing & Miscellaneous Parts: Rs 600–Rs 800
- Total Estimated Cost per Unit: Rs 3300–Rs 4100

With bulk production, optimized design, and local sourcing, the cost could be lowered further, making it an affordable automation solution for homes, offices, and event setups.